

ORIGINAL CONTRIBUTION

Prevalence and Clinical Relevance of Antiphospholipid Antibodies in Ischemic Stroke: A Systematic Review and Meta-Analysis

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BACKGROUND: Antiphospholipid antibodies (aPL) are associated with an increased risk of thrombosis. However, individual studies have reported conflicting findings regarding the prevalence of aPL in patients with ischemic stroke and their association with first or recurrent events.

METHODS: Systematic searches of PubMed and Embase through May 13, 2025 were conducted to identify case-control, cohort, or cross-sectional studies investigating 3 questions related to aPL seropositivity (defined as positivity for either immunoglobulin G/immunoglobulin M anticardiolipin antibody, immunoglobulin G/immunoglobulin M anti- β 2 glycoprotein I antibody, or a lupus anticoagulant): (1) the prevalence of aPL seropositivity among individuals with ischemic stroke; (2) the association between aPL seropositivity and the risk of first ischemic stroke; and (3) the association between aPL seropositivity and the risk of recurrent ischemic stroke. Random-effects models with inverse weighting were used to calculate pooled prevalences and odds ratios (OR) with 95% CIs. Risk of bias was assessed using the ROBINS-I tool.

RESULTS: A total of 52 studies were included (40 case-control, 12 cohort). The pooled prevalence of seropositivity for any aPL among patients with ischemic stroke (51 studies, 9438 patients) was 19.1% (95% CI, 15.4%–23.0%; $P=95.8\%$). aPL seropositivity was associated with higher odds of first ischemic stroke (43 studies, 19 097 patients; OR, 2.93 [95% CI, 2.31–3.73]; $P=68.7\%$). The strongest associations were for lupus anticoagulant (OR, 6.69 [95% CI, 2.94–15.2]; $P=92.6\%$) and immunoglobulin G anticardiolipin antibody (OR, 2.56 [95% CI, 1.96–3.35]; $P=92.5\%$). The odds of recurrent ischemic stroke among patients with versus without any aPL seropositivity were not significantly different (OR, 1.20 [95% CI, 0.87–1.67]; 9 studies, 2873 patients, $P=11.4\%$).

CONCLUSIONS: Seropositivity for any aPL was present in nearly 1-in-5 patients with ischemic stroke and was associated with increased odds of ischemic stroke at presentation.

GRAPHIC ABSTRACT: A graphic abstract is available for this article.

Key Words: endothelial cells ■ glycoprotein ■ ischemic stroke ■ prevalence ■ thrombosis

Antiphospholipid antibodies (aPL) are autoantibodies that bind to phospholipids and phospholipid-binding proteins, which lead to the activation of endothelial

cells, an overproduction of tissue factor, platelet aggregation, and a hypercoagulable state.^{1–3} aPL are usually identified through seropositivity for anticardiolipin

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This manuscript was sent to Harold P. Adams, Jr., Guest Editor, for review by expert referees, editorial decision, and final disposition.

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Supplemental Material is available at <https://www.ahajournals.org/doi/suppl/10.1161/STROKEAHA.126.055247>.

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Nonstandard Abbreviations and Acronyms

aCL	anticardiolipin antibody
APASS	Antiphospholipid Antibodies and Stroke Study
aPL	antiphospholipid antibody
APS	antiphospholipid antibody syndrome
aβ2GPI	anti-β2-glycoprotein I antibody
OR	odds ratio

antibody (aCL), aβ2GPI (anti-β2-glycoprotein I antibody), or a positive lupus anticoagulant test. The best recognized syndrome of aPL seropositivity is the thrombotic antiphospholipid syndrome (APS), characterized by persistent aPL seropositivity and either venous or arterial thrombosis events.^{4,5} Over the last decade, emerging evidence has suggested that aPL seropositivity may confer prognostic significance in a broader group of patients with cardiovascular conditions who may not fully satisfy traditional research criteria for APS.^{6–10}

Particularly considering ischemic stroke, the presence of a single positive aPL may represent a potential biomarker associated with excess risk of ischemic stroke, although findings have remained inconclusive across individual studies.^{11–16} For example, in the large prospective APASS (Antiphospholipid Antibodies and Stroke Study) of 1770 patients, the presence of aCL or a lupus anticoagulant among those with ischemic stroke did not predict recurrent thrombo-occlusive events, although many were treated with warfarin, which diminished their overall thrombotic risk.¹⁷ In contrast, other studies have demonstrated associations between aPL seropositivity and an increased risk of both first and recurrent stroke occurrence, further signifying the clinical uncertainty.^{18–20}

In this systematic review and meta-analysis of published studies, we aimed to assess the overall prevalence of aPL seropositivity among patients with ischemic stroke and evaluate the association between aPL seropositivity and both first and recurrent ischemic stroke.

METHODS

The authors declare that all supporting data are available within the article or its [Supplemental Material](#).

Systematic Search and Eligibility Criteria

This systematic review and meta-analysis was conducted and reported in accordance with the Preferred Reporting Items for Systematic Reviews and Meta-Analyses 2020 reporting guideline. This analysis was conducted in alignment with a prespecified protocol that was registered with the international prospective systematic review registry, PROSPERO

(CRD420251072953). A systematic search of both PubMed and Embase was conducted to identify studies of adult patients (without clinically diagnosed APS) in the English language investigating 1 of 3 questions: (1) the prevalence of aPL (or aPL subtypes) among individuals with ischemic stroke, (2) the association between aPL seropositivity and risk of first ischemic stroke, or (3) the association between aPL seropositivity and the risk of recurrent ischemic stroke. For the prevalence analysis, eligible study designs included individuals with stroke from case-control studies, prospective and retrospective cohort studies, and cross-sectional studies that reported the proportion of aPL seropositivity among patients with ischemic stroke. For the association between aPL seropositivity and first ischemic stroke, eligible studies were case-control studies (both population-based and hospital-based) and cohort studies that reported comparative data between patients with and without aPL seropositivity. For the association with recurrent ischemic stroke, eligible studies were cohort studies that reported recurrence data among patients with a prior ischemic stroke stratified by aPL status. The systematic search was conducted using terms related to aPL (eg, anticardiolipin, anti-β2 glycoprotein I, lupus anticoagulant, APS) combined with stroke-related terms (eg, ischemic stroke, stroke), with the full search strategy detailed in [Table S1](#). No restrictions were applied to study design in the initial search.

Included studies for prevalence estimates were required to report the proportion of aPL seropositivity among patients with ischemic stroke. Studies contributing to the association between aPL seropositivity and ischemic stroke were either cohort or case-control studies. All identified records were compiled within the systematic review manager Covidence (www.covidence.org), and all duplicate records were removed. Two independent reviewers (N.W. and S.B.) reviewed titles, abstracts, and relevant full-text articles to identify included studies. Identified records were excluded if they were case reports, small case series (<10 patients), non-English studies, or conference abstracts that had not yet undergone publication in a peer-reviewed journal. Data extraction was performed by 2 independent reviewers (N.W. and S.B.) and included study characteristics (author, year, study design, sample size), patient demographics (age, sex), aPL assay methods and positivity thresholds, and outcome data. For prevalence estimates, the number of aPL-seropositive patients and the total number of patients with ischemic stroke were abstracted directly from each study. For association analyses, raw event data were extracted and used to calculate odds ratios (ORs). A risk of bias assessment of all included cohort and case-control studies was performed utilizing the Cochrane ROBINS-I tool.²¹ The date of last database search was May 13, 2025 ([Table S1](#)).

Study Definitions

Seropositivity for aPL was defined as at least a single positive measurement for either lupus anticoagulant measured via a coagulation-based assay, immunoglobulin M (IgM)/immunoglobulin G (IgG) aβ2GPI, or IgM/IgG aCL above the normal assay limits among patients with ischemic stroke. To investigate a population of aPL seropositivity extending beyond the formal APS classifications, persistent positivity requiring repeated testing was not required. Specific dilutional cutoffs for seropositivity for each aPL and the assays used were defined

based on the definitions applied in each included study (Table; Table S2). Noncriteria antibodies (such as anti-prothrombin, anti-phosphatidylserine, anti-phosphatidylethanolamine, or annexin A5 resistance, among others²²) were not considered for this analysis.

Statistical Analysis

To estimate the proportion of patients with aPL seropositivity (including aPL subtypes), meta-analyses using random effects models with inverse variance weighting were applied. A Freeman-Tukey Double arcsine transformation method was used to generate proportions and the associated CIs of the estimates. To investigate the association between aPL seropositivity and first ischemic stroke, random-effects models generated by inverse variance weighting were used to calculate pooled ORs with 95% CIs. To compare the odds of recurrent ischemic stroke among those with versus without aPL seropositivity, random-effects models with inverse variance weighting were applied to calculate the pooled OR and the associated 95% CI. Where feasible, all meta-analyses were stratified by study design (case-control versus cohort) to assess the consistency of findings across study types. Two sensitivity analyses were planned for this study. First, the analysis was repeated with outlier studies excluded, defined as studies with CIs that did not overlap with the 95% CI of the pooled effect estimate. First, the analysis was repeated with outlier studies excluded, defined as studies with CIs that did not overlap with the 95% CI of the pooled effect estimate, to assess whether extreme values disproportionately influenced the pooled results.^{23,24} Second, the analysis was repeated only for case-control studies

which confirmed that aPL seropositivity was measured within 3 months of stroke presentation.

Between-study variance in each random-effects model was assessed with the τ^2 calculation, which was reported with all models. Model heterogeneity was assessed using the I^2 calculation, with $I^2 > 50\%$ being indicative of high heterogeneity. Publication bias was evaluated by analyzing funnel plots for asymmetry of included studies. All analyses were performed using R (version 4.1.0) in RStudio. Figures were prepared using R, BioRender, and Microsoft PowerPoint.

RESULTS

Among a total of 3413 unique records identified, 129 articles were assessed for full-text eligibility. Of these, 77 were excluded for the following reasons: 17% (n=22) for the wrong study population or outcome, 17% (n=22) for the wrong study design, 12% (n=15) for insufficient or nonextractable aPL-specific data, 5% (n=6) for unpublished conference abstracts, 3% (n=4) for duplicate records, and 7% (n=9) for other reasons. A total of 52 studies were included for analyses reported in this article (Figure S1).^{12–20,25–67} Of the included articles, 40 were case-control studies, and 12 were cohort studies. Of the included studies, 51 reported the proportion of aPL seropositivity among patients with ischemic stroke, contributing to the prevalence estimates. Among the 40 case-control studies, the majority (n=35, 88%) were hospital-based, enrolling stroke patients admitted

Table. Summary of LA Assays and aCL, aB2GP Cutoffs Applied Across Included Studies*

aPL		Assays	Cutoff definition/range	Comments
Lupus anticoagulant		aPTT and KCT initially	Positive if screen/confirm ratio >99th percentile of plasma of healthy controls	...
		DRVVT and StacLOT-LA used for confirmation		
a β_2 GPI	IgG	ELISA	Positivity: >16–23 U/mL	One study used the cutoff as the 99th percentile of the population with definite antiphospholipid syndrome.
			>90th–99th percentile of control group or healthy population	
	IgM	ELISA	Positivity: >11–20 U/mL	...
			>90th–99th percentile of control group or healthy population	
aCL	IgG	ELISA	Positivity: >10–30 GPL U	Some studies reported positivity as above 2–5 standard deviations above the mean of the control group.
			>90th–99th percentile of control group or healthy population	
			Low: 10–20 GPL	
			Moderate: 21–40 GPL	
	IgM	ELISA	High:>40 GPL	For the positivity ranges, a few studies reported higher upper-limit cutoffs of 60, 80, or 100 U to define moderate and high positivity.
			Positivity: >10–20 MPL or >90th–99th percentile of control group or healthy population	
			Low: 10–20 MPL U	
			Moderate: 21–40 MPL U	
			High:>40 U	

aCL indicates anticardiolipin antibody; aPL, antiphospholipid; aPTT, activated partial thromboplastin time; a β_2 GPI, anti- β_2 -glycoprotein I antibody; DRVVT, dilute Russell viper venom time; GPL, IgG phospholipid units; IgG, immunoglobulin G; IgM, immunoglobulin M; ISTH, International Society on Thrombosis; KCT, kaolin clotting time; LA, lupus anticoagulant; MPL, IgM phospholipid units; and StacLOT-LA, silica clotting time assay for lupus anticoagulant detection.

*See Table S2 for details.

to 1 or multiple institutions, and only 4 (10.0%) were population-based. Among the 12 cohort studies examining recurrent stroke, 11 (92%) were hospital-based, and 1 (8%) was population-based. Forty-three out of 52 studies, encompassing 19 097 patients, included individuals with a first-ever ischemic stroke event, while the remaining 9 studies, representing 2873 patients with a history of prior stroke, were included in the analysis of aPL seropositivity and recurrent stroke risk (Table S3). Within the entire cohort, the mean age was 54.0 ± 14.2 years and 47.9% were female. Table S4 contains specifications of the ischemic stroke and control groups for each included study.

For assessment of aPL prevalence, 51 articles were included, encompassing a total of 9438 patients. Among included studies, the pooled prevalence of seropositivity for any aPL among patients with ischemic stroke was 19.1% (95% CI, 15.4%–23.0%; $P=95.8\%$; Figure 1; Figure S2). When stratified by study design, the pooled prevalence of aPL seropositivity was 18.8% (95% CI, 14.4%–23.7%) among case-control studies and 20.1% (95% CI, 14.2%–26.7%) among cohort studies. In a sensitivity analysis performed after removal of outlier studies, the pooled prevalence was 20.5% (95% CI, 17.9%–23.3%; $P=78.7\%$), demonstrating that the primary findings were not driven by a small subset of studies with extreme values and that the magnitude and direction of the estimate remained consistent (Figure S3). Similar findings were observed in a sensitivity analysis restricted to studies that specified that aPL were assayed within 3 months of ischemic stroke, with a pooled prevalence of 20.6% (95% CI, 12.3%–30.3%; Figure S4). Seropositivity for aPL subtypes including a lupus anticoagulant, IgG a β 2GPI, IgM a β 2GPI, IgG aCL, and IgM aCL were 12.0% (95% CI, 6.75%–18.4%), 10.3% (95% CI, 5.15%–16.9%), 8.92% (95% CI, 2.94%–17.6%), 14.5% (95% CI, 11.3%–18.0%), and 12.1% (95% CI, 8.47%–16.2%), respectively (Figures S5 through S9).

A total of 43 articles encompassing 19 097 patients (6367 patients with stroke and 12 730 patients without

stroke) were included in the analysis investigating the association between seropositivity for aPL and first ischemic stroke. Patients seropositive for aPL had an overall higher odds for first ischemic stroke (pooled OR, 2.93 [95% CI, 2.31–3.73]; $P=68.7\%$), with consistent results among cohort studies (pooled OR, 2.25 [95% CI, 1.52–3.33]; $P=33.3\%$) and case-control studies (pooled OR, 3.04 [95% CI, 2.31–4.00]; $P=70.5\%$, Figure 2). Similar findings were demonstrated in a sensitivity analysis following the removal of outlier studies, with an improvement in model heterogeneity (pooled OR, 2.69 [95% CI, 2.23–3.24]; $P=41.9\%$; Figure S10). In a sensitivity analysis restricted to case-control studies that specified aPL were assayed within 3 months of ischemic stroke, a similarly increased odds for first ischemic stroke was observed with aPL seropositivity (pooled OR, 3.40 [95% CI, 2.24–5.17]; $P=55.3\%$; Figure S11).

Figure 3 depicts the summary findings of the association between aPL seropositivity and first ischemic stroke stratified by aPL subtype. Patients with a positive test for lupus anticoagulant were at 6.69-fold higher odds of ischemic stroke (pooled OR, 6.69 [95% CI, 2.94–15.2]; $P=75.3\%$; Figure S12). Additionally, those seropositive for IgG a β 2GPI also had higher odds of first ischemic stroke (pooled OR, 2.12 [95% CI, 1.19–3.75]; $P=55.4\%$), although findings for IgM a β 2GPI showed no association (pooled OR of 1.73 [95% CI, 0.51–5.89]; $P=77.9\%$; Figures S13 and S14). Patients seropositive for IgG aCL (pooled OR, 2.56 [95% CI, 1.96–3.35]; $P=62.3\%$) and IgM aCL (pooled OR, 1.99 [95% CI, 1.40–2.83]; $P=73.2\%$) also demonstrated increased odds of ischemic stroke (Figures S15 and S16).

A total of 9 studies representing 2873 patients with a history of prior stroke were included for analysis of aPL seropositivity and the risk of recurrent stroke. All included studies were cohort studies, comprising 8 prospective and 1 retrospective cohort study, with follow-up durations ranging from 12 months to 8.9 years. Studies varied in population characteristics, including age range (young adults to elderly), stroke subtype (cryptogenic

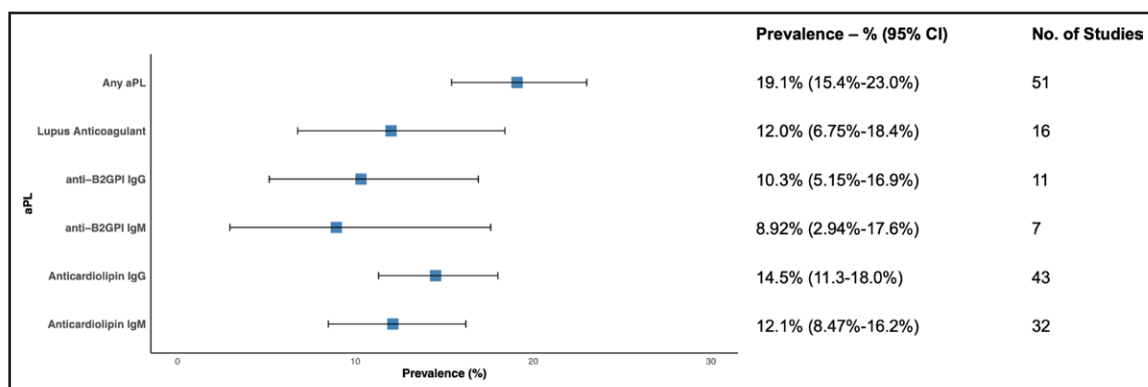


Figure 1. Prevalence of antiphospholipid (aPL) antibodies in ischemic stroke by aPL subtype (51 studies, 39 case-control studies, and 12 cohort studies).

IgG indicates immunoglobulin G; and IgM, immunoglobulin M.

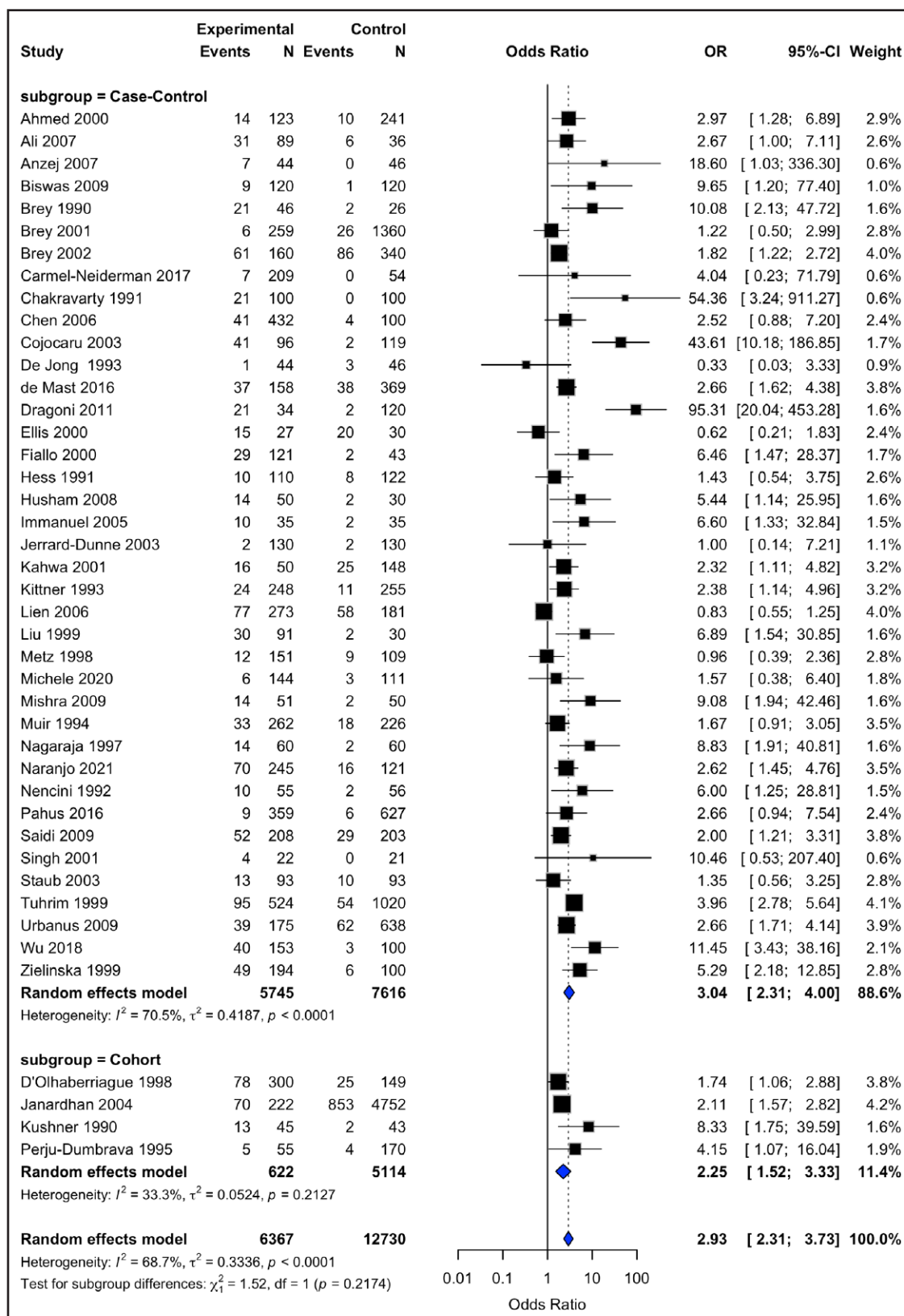


Figure 2. Pooled odds of first ischemic stroke by antiphospholipid (aPL) seropositivity status.

Analysis stratified by study type, including either cohort or case-control study (43 studies: 39 case-control, 4 cohort). OR indicates odds ratio.

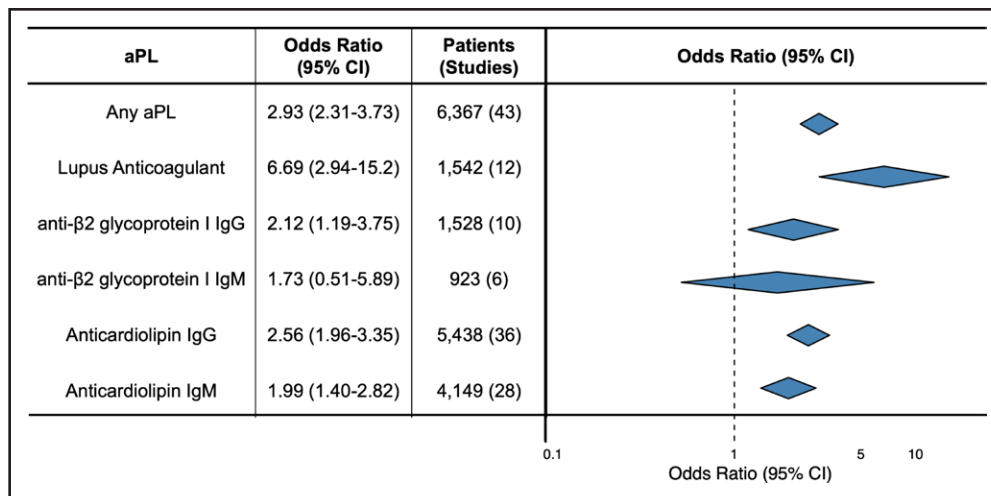


Figure 3. Pooled odds of first ischemic stroke by antiphospholipid (aPL) subtype seropositivity status for all aPL subsets (43 studies: 39 case-control, 4 cohort).

IgG indicates immunoglobulin G; and IgM, immunoglobulin M.

versus unselected), and whether aPL were measured once or serially. In pooled analysis, there was no significant difference in the odds of recurrent stroke among patients with versus without aPL seropositivity (pooled OR, 1.20 [95% CI, 0.87–1.67]; $P=11.4\%$; Figure 4). Individual studies did not report adjusted findings for cotreatments poststroke, including statins, antiplatelet agents, and anticoagulants. Among studies that only included patients with a mean age <50 years or those with cryptogenic stroke, there was no association of aPL with odds of recurrent stroke (N=433 patients; pooled OR, 1.11 [95% CI, 0.60–2.06]; $P=0.00\%$; Figure S17).

Table S5 depicts the summary results for the risk of bias analysis for the 52 included studies in this analysis. In total, 45 studies (86.5%) were at an overall moderate risk of bias and 7 studies (13.5%) at a serious risk

of bias. Among the 40 case-control studies, 35 (87.5%) were at moderate risk of bias, and 5 (12.5%) were at serious risk of bias. Among the 12 cohort studies, 10 (83.3%) were at moderate risk of bias, and 2 (16.7%) were at serious risk of bias. Assessment of individual risk of bias components can be found in Table S5. Figures S18 through S20 illustrate funnel plots for assessment of publication bias among the 3 primary analyses performed.

DISCUSSION

Approximately 1 in 5 patients with ischemic stroke exhibited seropositivity for at least 1 type of aPL, with the highest prevalence seen for IgG/IgM aCL and lupus anticoagulant. Patients seropositive for at least 1 aPL had nearly 3-fold higher odds of first ischemic stroke,

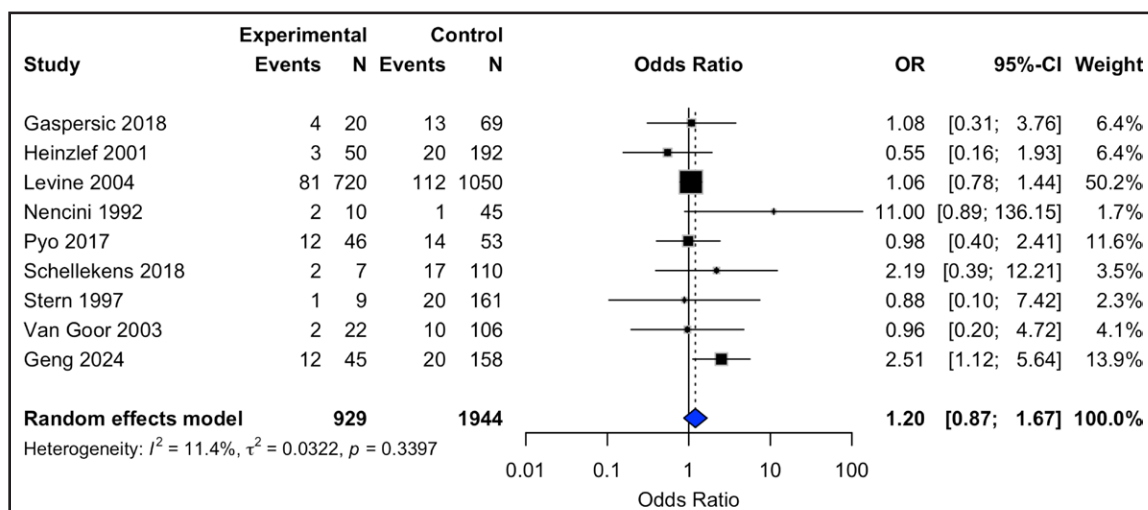


Figure 4. Pooled odds of recurrent ischemic stroke among patients with antiphospholipid (aPL) seropositivity vs those without aPL seropositivity (9 studies, all cohort studies). OR indicates odds ratio.

with the strongest associations observed for lupus anticoagulant and IgG aCL. Importantly, all aPL subtypes demonstrated statistically significant associations with ischemic stroke except for IgM anti- β 2GPI, a finding that may reflect greater pathogenicity of IgG isotypes compared with IgM. Nevertheless, this finding should be interpreted cautiously given the small number of patients contributing to analyses related to IgM, limiting the statistical power to detect an association. However, aPL seropositivity was not associated with significantly increased odds of recurrent ischemic stroke in patients with a prior stroke, even when restricted to younger patients or those with cryptogenic stroke ([Graphical Abstract](#)).

These findings offer several important insights into the complex relationship between aPL and ischemic stroke. Notably, the association of aPL seropositivity with first ischemic stroke, particularly lupus anticoagulant and IgG aCL, along with consistent associations both in case-control and cohort studies, suggests a potential role for aPL in the pathogenesis or unmasking of thrombogenic risk. This observation points to a biologically relevant subset of patients in whom single-time aPL seropositivity in the context of ischemic stroke, though not confirmed over time and as such insufficient to meet APS research classification criteria,^{5,8} nonetheless reflects a prothrombotic state.¹ It should be clarified that the APS classification criteria, such as the Sapporo and EULAR criteria, were designed for research standardization rather than defining the full spectrum of clinical risk. This meta-analysis of epidemiological studies was not designed to, and cannot, directly support or refute the Sapporo or ACR/EULAR research criteria for APS. However, our findings plausibly indicate that single-time aPL seropositivity may be associated with excess risk of stroke. To what extent this associated risk reflects persistent positivity versus some contribution from individuals with transient positivity cannot be ascertained from our study, as no large-scale source investigations with serial aPL testing were identified.

Although single-time positive aPL is insufficient to meet traditional research classification criteria, it may reflect population-level signal for excess risk. It remains to be determined whether such a signal is directly pathogenic (causal) or may also represent an epiphenomenon related to immune activation in the setting of acute ischemic injury or systemic inflammation. In addition, variability in assay methodology, differences in seropositivity thresholds, and the absence of repeat confirmatory testing at ≥ 12 weeks may contribute to false-positive results and misclassification of clinically meaningful aPL status. Accordingly, the associations observed in this meta-analysis, while plausible and possible for a causal mechanism, should not be interpreted as definitive proof of causality. Rather, aPL seropositivity likely represents a spectrum, ranging from true pathogenic contributors in some individuals to markers of underlying vascular or

inflammatory states in others.¹⁰ In these individuals, ischemic stroke may represent the first clinical manifestation of an underlying prothrombotic tendency, implicating aPL as context-dependent modulators of thrombotic risk that tips the hemostatic balance under specific pathophysiologic conditions. Alternatively, aPL may function as risk markers without direct pathogenicity, or as epiphenomena of the immune response to ischemic brain injury. However, multiple lines of evidence—including mechanistic studies, animal models, and clinical associations in cohort studies—support their pathogenicity, particularly for a β 2GPI, in arterial thrombosis through endothelial activation, complement engagement, and synergistic interaction with other vascular risk factors.^{10,68,69} Notably, emerging evidence suggests that the association between aPL seropositivity and excess thromboembolic risk may extend beyond those with formally confirmed persistent positivity, with consistent associations observed across a range of cardiovascular conditions, including in studies where repeat longitudinal testing was not systematically performed.¹⁰

We did not observe a significant association between aPL seropositivity and recurrent stroke, consistent with a prior meta-analysis of 8 studies (2510 patients, 844 aPL-seropositive; RR, 1.41 [95% CI, 0.91–2.17]).²¹ One likely explanation is that the prothrombotic risk of aPL evident at the index stroke is attenuated by post-stroke treatments like anticoagulants, antiplatelet agents, or statins, which reduce subsequent recurrence and obscure the underlying biological risk signal. It is also possible that anticoagulation is applied differentially in aPL-associated stroke, as significant variations in antithrombotic management have been reported across clinical settings.⁷⁰ Additionally, variation in patient demographics, stroke subtypes, follow-up duration, and treatment contributed to heterogeneity across studies. Since recurrent stroke is relatively uncommon and many studies were underpowered, modest but meaningful associations may go undetected (ie, type II error). This pattern of attenuated association at recurrence is not unique to aPL and has been described for other thrombophilic conditions—most notably factor V Leiden, where the association with first venous thromboembolism similarly exceeds that observed for recurrent events.⁷¹ This phenomenon is explained, in part, by index event bias, which occurs when analyses are restricted to patients who have already experienced an initial thrombotic event, distorting the relationship between risk factors and recurrence. Among patients with a first ischemic stroke, individuals without aPL antibodies are more likely to have experienced their initial event because of other strong vascular risk factors, attenuating the apparent association between aPL antibodies and recurrent stroke.⁷² Future individual cohort studies should consider implementing strategies such as inverse probability weighting or multivariable adjustment for other stroke risk factors to account for

this bias. Finally, limited assessment of aPL pathogenicity (eg, persistence, high titer, multiple antibodies) suggests that some seropositive cases may reflect transient or low-risk states. It is conceivable that a biological gradient exists whereby patients with higher aPL titers, multiple antibody positivity, and persistent seropositivity—particularly in the absence of aggressive secondary prevention strategies—may face a greater risk of recurrent ischemic stroke.¹⁰ Identifying aPL profiles that reflect true risk versus biological noise remains challenging. These findings should therefore not be interpreted as definitive evidence against the clinical relevance of aPL as predictors of future stroke, including in patients with established APS.

This study has several limitations. First, this study should be interpreted in the context of the variable methodological quality of the included studies, which may limit the precision of the pooled estimates. However, results were directionally robust over a variety of sensitivity analyses. As such, although there is imprecision in the magnitude of effect, our findings provide confirmatory evidence on the excess risk of stroke with aPL seropositivity. Moreover, notable heterogeneity was observed across the studies included in this analysis. This was likely due to variations in assays and methodology applied for aPL measurement, differences in seropositivity thresholds, and overall heterogeneity in the case mix among individual studies. Given the high degree of heterogeneity in our prevalence estimates, it is likely that our analysis captured both populations that enriched the estimates through higher prevalence than average and those that yielded lower prevalence estimates than average. Additionally, antibody levels and titer-specific risk estimates were not consistently reported across studies, precluding stratified analyses based on aPL level. Nonetheless, our sensitivity analyses excluding outlier studies demonstrated consistent findings with reductions in heterogeneity, suggesting that the observed associations were not driven by a small subset of studies. Second, there were likely differences in the timing of antibody measurement relative to stroke onset among included studies, which may have introduced bias into the analysis. However, findings were relatively consistent among studies that included information on the timing of measurements. Third, it remains possible that other factors, including time-varying vascular risk factors, comorbid autoimmune disease, or differences in treatment, may have influenced the presented results in ways that were not adequately controlled in the individual studies and hence the pooled analysis. These modifiers may have alleviated the differences observed in the pooled estimates. Fourth, the funnel plot for the prevalence analysis demonstrated asymmetry suggestive of potential publication bias, raising the possibility that the pooled aPL prevalence estimate of 19.1% may represent an overestimate of the true population prevalence. Smaller studies with higher prevalence estimates may be disproportionately represented in the published literature,

and this should be considered when interpreting these findings. Finally, differences in ischemic stroke subtypes (eg, cardioembolic, large vessel, cryptogenic), patient demographics (age, vascular risk factors, autoimmune comorbidities), and concomitant anticoagulation utilization were often incompletely reported, limiting our ability to perform meaningful subgroup analyses. In hospitalized cohorts with clinician-initiated testing, aPL assessment is likely influenced by patient age and comorbidities, as clinicians may be less inclined to test older individuals with conventional vascular risk factors—potentially underestimating true aPL prevalence. Moreover, there is a lack of robust data on social determinants of health, including race and ethnicity, socioeconomic status, and health care access, which may influence both the likelihood of aPL testing and subsequent management decisions. Finally, the majority of included studies were published before 2010. As aPL assay methodology, seropositivity thresholds, and clinical testing recommendations have evolved substantially in the intervening period, the results should be interpreted through this lens.

CONCLUSIONS

Findings from this comprehensive systematic review and meta-analysis indicate that aPL are frequently detected among patients with ischemic stroke and are associated with higher odds of first ischemic stroke. By contrast, aPL seropositivity was not significantly associated with increased odds of recurrent stroke, which may be due to increased utilization of poststroke antithrombotic therapy or other reasons. Future prospective studies with standardized aPL testing protocols and stratification by stroke subtype can further clarify the prognostic implications of aPL seropositivity and a potential modifying role in the effectiveness of secondary prevention therapies.

ARTICLE INFORMATION

Received January 6, 2026; final revision received July 1, 2026; accepted August 14, 2026.

Presented in part at the American Heart Association Scientific Sessions, New Orleans, LA, November 10, 2025.

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Sources of Funding

None.

Disclosures

Dr Weber reports compensation from Journal of Nuclear Cardiology for other services; compensation from Oruka for consultant services; employment by University of Texas Southwestern Medical Center; compensation from Bristol Myers Squibb Company for other services; and compensation from Novo Nordisk for consultant services. Dr Kanthi reports employment by the National Institutes of Health. Dr Secemsky reports compensation from Inari Medical Inc for consultant services; stock options in Thrombolex; compensation from Cardiovascular Systems Inc. for consultant services; compensation from Philips for consultant services; compensation from AngioDynamics Inc for consultant services; compensation from Infrar for consultant services; compensation from Abbott Vascular for consultant services; compensation from Janssen Biotech for consultant services; compensation from ZOLL Medical Corporation for consultant services; compensation from VentureMed Group for consultant services; compensation from Cook Incorporated for consultant services; stock options in endovascular engineering; compensation from Medtronic for consultant services; stock options in Inquis; compensation from W. L. Gore & Associates Inc for consultant services; compensation from BARD for consultant services; compensation from Rampart for consultant services; compensation from Teleflex LLC for consultant services; stock options in Innova; compensation from Thrombolex for consultant services; compensation from Terumo for consultant services; compensation from Inari Medical Inc for consultant services; compensation from ShockWave Medical Inc for consultant services; compensation from Bristol Myers Squibb Company for consultant services; compensation from Bayer for consultant services; employment by Beth Israel Deaconess Medical Center; compensation from endovascular engineering for consultant services; compensation from BOSTON SCIENTIFIC CORPORATION for consultant services; compensation from RapidAI for consultant services; compensation from Penumbra Inc for consultant services; and compensation from Siemens for consultant services. Dr Connors reports compensation from Bristol Myers Squibb Company for consultant services; compensation from Janssen Pharmaceuticals Inc for data and safety monitoring services; compensation from Novartis for consultant services; compensation from Regeneron Pharmaceuticals Inc for consultant services; compensation from Perosphere for consultant services; compensation from Anthos for consultant services; compensation from Pfizer for consultant services; compensation from Bayer for consultant services; compensation from Alexion Pharmaceuticals Inc for data and safety monitoring services; and compensation from Cerus Corporation for data and safety monitoring services. Dr Barnes reports compensation from Anthos for consultant services; service as President-Elect for Anticoagulation Forum; compensation from Pfizer for consultant services; employment by University Michigan; compensation from Janssen Biotech Inc for consultant services; compensation from Bayer for consultant services; grants from BOSTON SCIENTIFIC CORPORATION; compensation from Bristol-Myers Squibb for consultant services; and compensation from Pfizer for consultant services. Dr Uljon reports compensation from Werfen USA LLC for consultant services and grants from Grifols Biologicals Inc. Dr Goldhaber reports compensation from Boston Scientific Corporation for consultant services; grants from Janssen Pharmaceuticals Inc; grants from National Heart and Lung Institute; grants from Boston Scientific Corporation; grants from Bristol Myers Squibb Company; grants from Bayer; and grants from Pfizer. Dr Weitz reports compensation from Daiichi Sankyo Company for consultant services; compensation from Janssen Global Services, LLC for consultant services; compensation from Ionis Pharmaceuticals for consultant services; compensation from Alnylam Pharmaceuticals for consultant services; compensation from Bristol Myers Squibb Company for consultant services; compensation from Boehringer Ingelheim for consultant services; compensation from Pfizer for consultant services; compensation from Ionis for consultant services; compensation from Anthos for consultant services; compensation from VarmX Pharmaceuticals for

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Supplemental Material

Tables S1–S5

Figures S1–S20

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